

Express.js Full Roadmap (Basic to Advanced – Industry Level)

Phase 1: Express Basics

- What is Express.js and why it is used
- Express vs Node.js HTTP module
- Creating first Express server
- Routing basics (GET, POST, PUT, DELETE)
- Request and Response objects

Phase 2: Middleware & Core Concepts

- What is middleware
- Types of middleware
- Built-in middleware
- Custom middleware
- Error handling middleware

Phase 3: REST API Development

- REST principles
- Designing RESTful APIs
- Controllers and routes separation
- CRUD operations
- HTTP status codes best practices

Phase 4: Database Integration

- MongoDB with Mongoose
- Schemas and Models
- CRUD operations
- Relationships and pagination
- SQL integration overview

Phase 5: Authentication & Authorization

- Authentication vs Authorization
- JWT based authentication
- Role-based access control
- Password hashing with bcrypt
- Security best practices

Phase 6: Advanced Express

- Async/Await handling
- Centralized error handling
- File uploads with Multer
- Request validation
- API versioning

Phase 7: Performance & Scalability

- Performance optimization
- Caching concepts
- Compression
- Load balancing basics
- Logging and monitoring

Phase 8: Testing & Documentation

- Unit testing
- Integration testing
- Jest and Supertest
- Swagger API documentation
- Postman collections

Phase 9: Deployment & DevOps

- Environment configuration
- PM2 process manager
- Docker basics
- CI/CD overview
- Cloud deployment

Phase 10: Real World Projects

- User Management System
- Authentication Service
- E-commerce Backend
- Learning Management System
- Job Portal API